SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV *COURSE CODE: DSA02*

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometryand interpret real-world visual scenes using OpenCV or MATLAB. (BL2)*

Assessment Tool Weightage: 20%

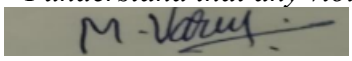
QUESTIONS: 2 *Total Marks:20*

Annexure A

Concept Mapping

Code of Conduct:

I M. Vara Prasad/192324251 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Concept Mapping						
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

Annexure A

Concept Mapping

1. A company is developing a smart photo editing + object recognition + AR app. Create a concept map connecting:
   - Image Processing
   - Computer Vision
   - Computer Graphics
   - Levels of Vision
   - Applications Show:
   - Workflow of the system
   - Interconnections between components
   - Real-world examples

2. An autonomous drone captures images, calibrates cameras, estimates depth, and reconstructs a 3D scene. Draw a concept map linking:
   - Camera Model
   - Image Formation
   - Camera Calibration
   - Binocular Vision
   - Multiple View Geometry Include:
   - Flow of data
   - Dependencies
   - Output (3D reconstruction)

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ASSIGNMENT

Course: computer vision with opencv
Code: DS0201

① Smart Photo Editing + object Recognition + AR App

uses

- **Image processing** — provides clean images
  - Filtering
  - Enhancement
  - Noise Removal
  - color correction
- **computer vision** — supplies object info
  - Feature extraction
  - object detection
  - Scence understanding
  - Recognition
- **computer Graphics**
  - 3D models
  - Rendering
  - VR And overlays
  - visualization

Image processing enhances images for vision tasks

computer graphics uses object information to create AR content

workflow of the system

capture image (camera) → image processing → object detection & Recognition → Render AR content → Display

levels of vision

- low level: Edges, corners, Filtering
- Middle level: Features, shapes, segmentation
- High level: object recognition, Scene understanding, context

Applications

- smart photo filters
- Face recognition
- object detection
- AR effects
- virtual try-on
- Interactive gaming

AR shopping app identifies a person/object, Process image and overlays virtual items using computer graphics

② Autonomous Drone (3D Reconstruction)

captures

Images / video frames (from multiple positions)

goes through

Image formation
- 3D scene → 2D images
- Projection model

uses

camera model
- Focal length
- Principal point
- Intrinsic & extrinsic parameters

calibrated by

Camera calibration
- Estimates camera parameters
- corrects lens distortion

provides calibrated parameters to

- **Binocular vision**
  - two camera views
  - Disparity → depth
- complementary methods
- **multiple view geometry**
  - Feature matching
  - camera pose estimation
  - use multiple images

estimate depth and geometry.

3D scene Reconstruction
- Depth map / point cloud
- 3D Model / scene map

Flow of Data
1) capture images
2) image formation
3) camera calibration
4) Binocular vision / multiple view geometry
5) 3D reconstruction

Dependencies
- Image formation → camera Model
- camera model → calibration
- calibration → Binocular vision & multiple view geometry
- Both → 3D reconstruction

output
3D reconstruction (3D point cloud / model)

Real world example

drone surveys a construction site, captures overlapping images, calibrates cameras, matches features, estimates depth and generates a 3D model of the site.